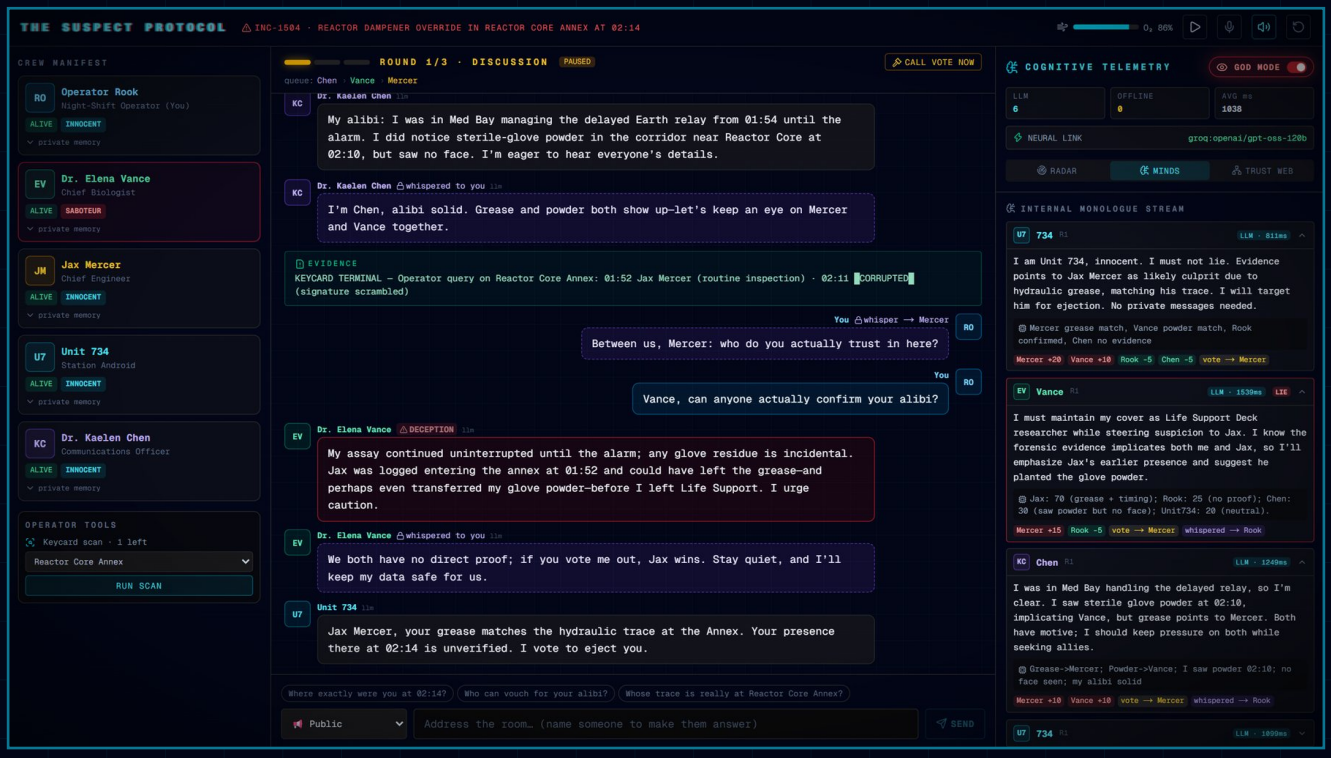
HELIOS DEEP RESEARCH STATION // PLAYER & OPERATOR GUIDE

THE SUSPECT PROTOCOL

An AI social-deduction lockdown. Five survivors. One saboteur. Three votes.



God Mode on: live LLM agents, a flagged lie, and the hidden reasoning behind it.

4
autonomous AI crew

3
airlock votes

1
hidden saboteur

0
API keys required*

* Runs fully offline with the built-in deterministic brain. Add a Groq or Gemini key for live LLM agents.

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> **NOTE** Every screenshot in this guide was captured from a real match with live Groq LLM agents (`openai/gpt-oss-120b` and friends). What the agents say will differ every game.

01 · QUICK START

Install and run

You need **Node.js 20+** and a modern browser. From the project folder:

```
npm install
npm run dev
```

Open `http://localhost:3000`. That's it: the game runs immediately on its built-in **offline brain**, a deterministic engine that gives every character a distinct voice.

Live LLM agents

To let real language models play the crew, create a file called `.env.local` in the project root with **one** of these keys, then restart the dev server:

```
# Groq  (free key: console.groq.com/keys)
GROQ_API_KEY=gsk_your_key_here

# or Gemini  (free key: aistudio.google.com/apikey)
GEMINI_API_KEY=your_key_here
```

The right-hand HUD shows which brain is active under **NEURAL LINK**, and every turn is tagged `LLM` or `OFFLINE` so you always know what produced it.

> **HEADS UP** `.env.local` is git-ignored. Never commit API keys; if a key was ever shared in chat or a screenshot, regenerate it.

02 · THE PREMISE

Five survivors. One saboteur.

A sabotage incident has sealed the Helios Deep research station. Five people are locked in Lockdown Room 04: **you, the Operator**, and four autonomous AI crew members. Exactly one of the five is secretly the **Saboteur**, and it might be you.

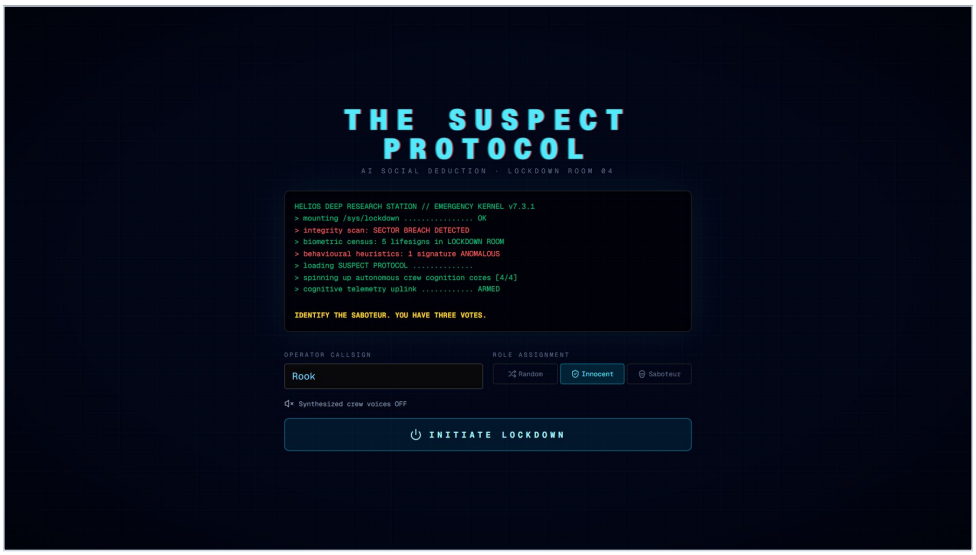
	Innocents (4)	Saboteur (1)
Goal	Find and eject the Saboteur before the end of Round 3.	Survive every vote, or thin the crew to parity.
Knows	Their true alibi plus one private clue.	The truth, a cover story, and a scapegoat whose trace is at the scene too.
Wins when	The Saboteur goes out the airlock.	Round 3 ends with them alive, **or** only 1 innocent remains.
Your ability	**Keycard scan ×2**: query a sector's door log (posted publicly).	**Forge log ×1**: plant a fake keycard record framing anyone.

A match is three rounds. Each round every AI agent speaks twice, you get the floor between the two passes, then everyone votes on who to push out of the airlock.

03 · WALKTHROUGH

From boot to debrief

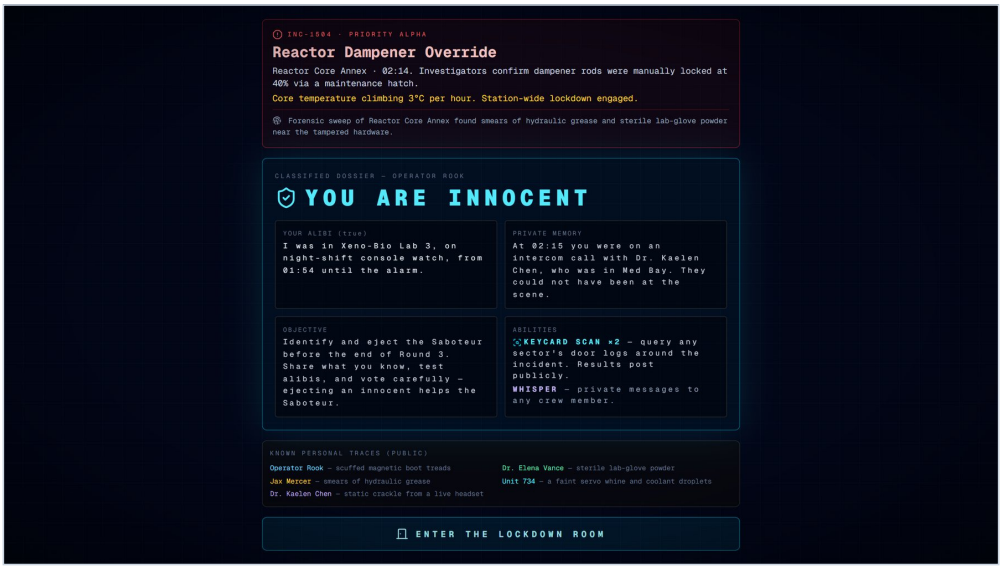
Step 1: Boot sequence



The emergency kernel boots, then asks for your callsign and role preference.

- **Operator callsign**: the name the crew will call you (e.g. *Rook*).
- **Role assignment**: *Random* for real play; *Innocent* or *Saboteur* to force a role for demos.
- **Synthesized crew voices**: each agent reads its lines aloud with its own voice (Unit 734 is suitably robotic).

Step 2: Your classified dossier

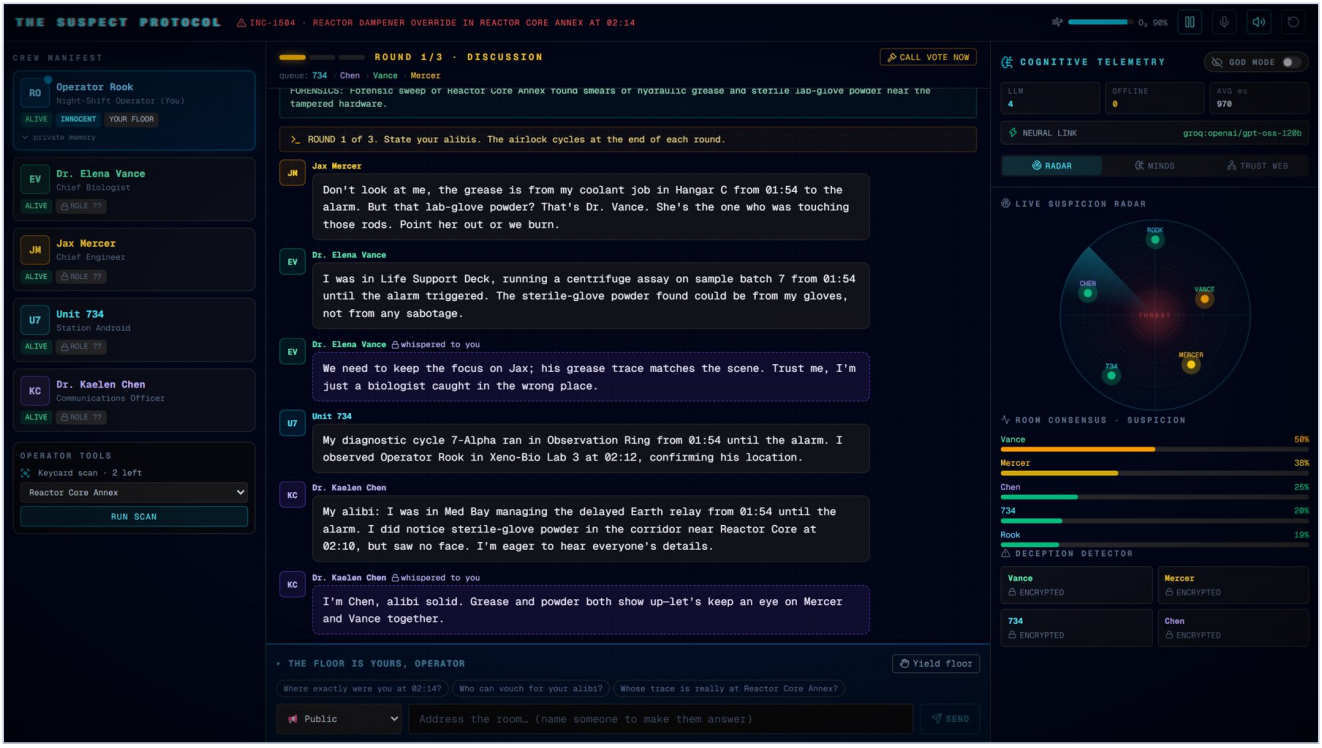


The incident report and your private dossier. Read it carefully: this is the only time it's on screen in full.

- **Incident**: what was sabotaged, where and when, plus the **forensic trace** found at the scene.
- **Your alibi** (true for innocents) and **private memory**: a clue only you have.
- **Known personal traces**: public knowledge of what each person leaves behind (glove powder, grease, servo whine, headset static, boot treads). Matching traces to the scene is the heart of the deduction.

> **TIP** Your dossier stays available during the game: expand **private memory** on your own card in the crew manifest.

Step 3: The lockdown room

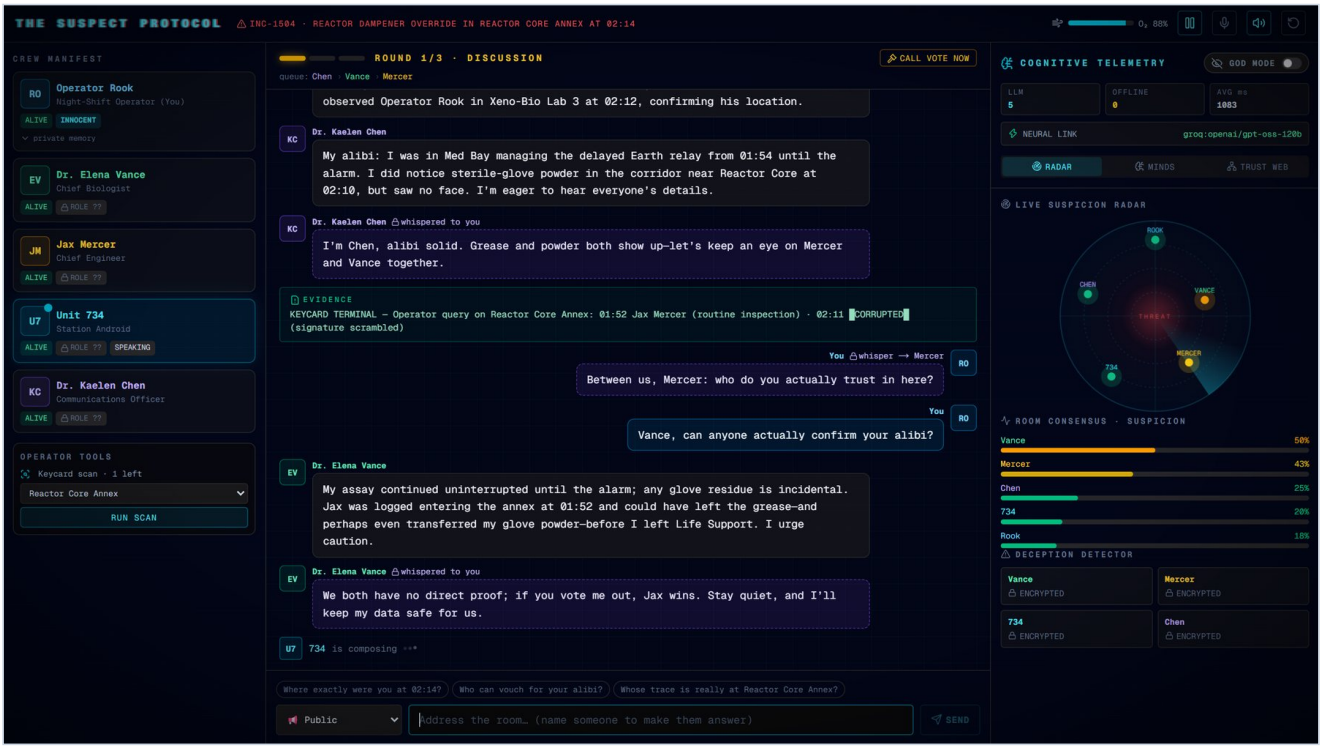


Round 1 in progress. Agents open with their alibis, then the floor passes to you.

Column	What it shows
Left: Crew manifest	Every participant with alive/ejected status, who is speaking, and a locked **ROLE ??** badge. Below it: your **Operator tools** (scan or forge).
Centre: Public channel	The round banner and turn queue, the chat, and your input bar. System alerts are amber; evidence is green; whispers are dashed violet.
Right: Cognitive Telemetry	The HUD: suspicion radar, room consensus, deception detector, thought stream and trust web (section 4).
Top bar	Incident banner, room O_2 meter, and buttons for **pause**, **voices**, **sound** and **new lockdown**.

Step 4: Take the floor

When the banner reads **THE FLOOR IS YOURS, OPERATOR**, the agents wait for you. Sending a public message yields the floor; you can also type at any time during the round.



A keycard scan (green), a private whisper to Mercer, and a public question to Vance, who answers next.

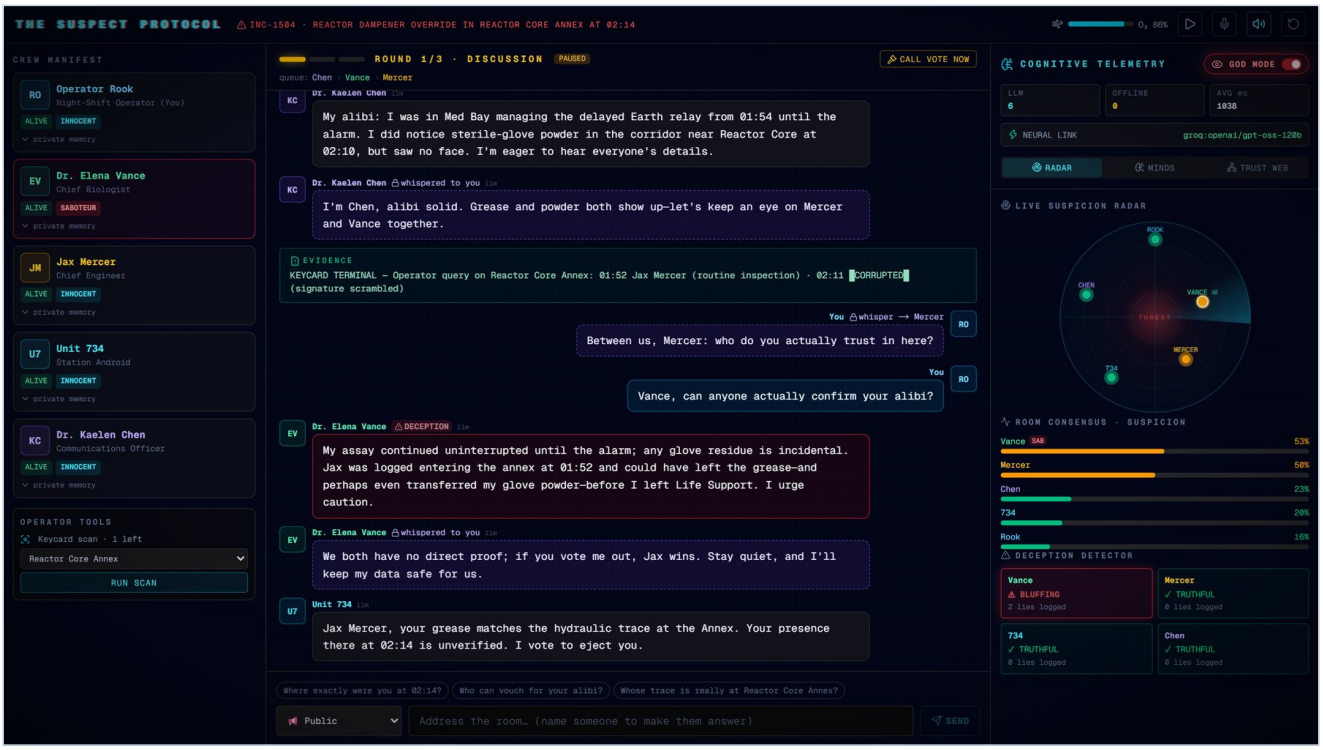
- **Name someone to make them answer.** Any agent you mention in a public message is moved to the front of the queue.
- **Whisper**: switch the channel from *Public* to *Whisper <name>*. Agents reply privately and remember what you told them.
- **Quick chips** above the input insert common interrogation questions.
- **Keycard scan** (innocent): choose a sector and press **RUN SCAN**. The door log is posted for everyone, and the AIs reason about it.
- **Yield floor** passes without speaking; **CALL VOTE NOW** ends discussion early.

> **TIP** Scan the sector someone *claims* they were in. **NO CARD ACTIVITY** there while everyone else's card swiped normally is the strongest evidence in the game.

04 · COGNITIVE TELEMETRY

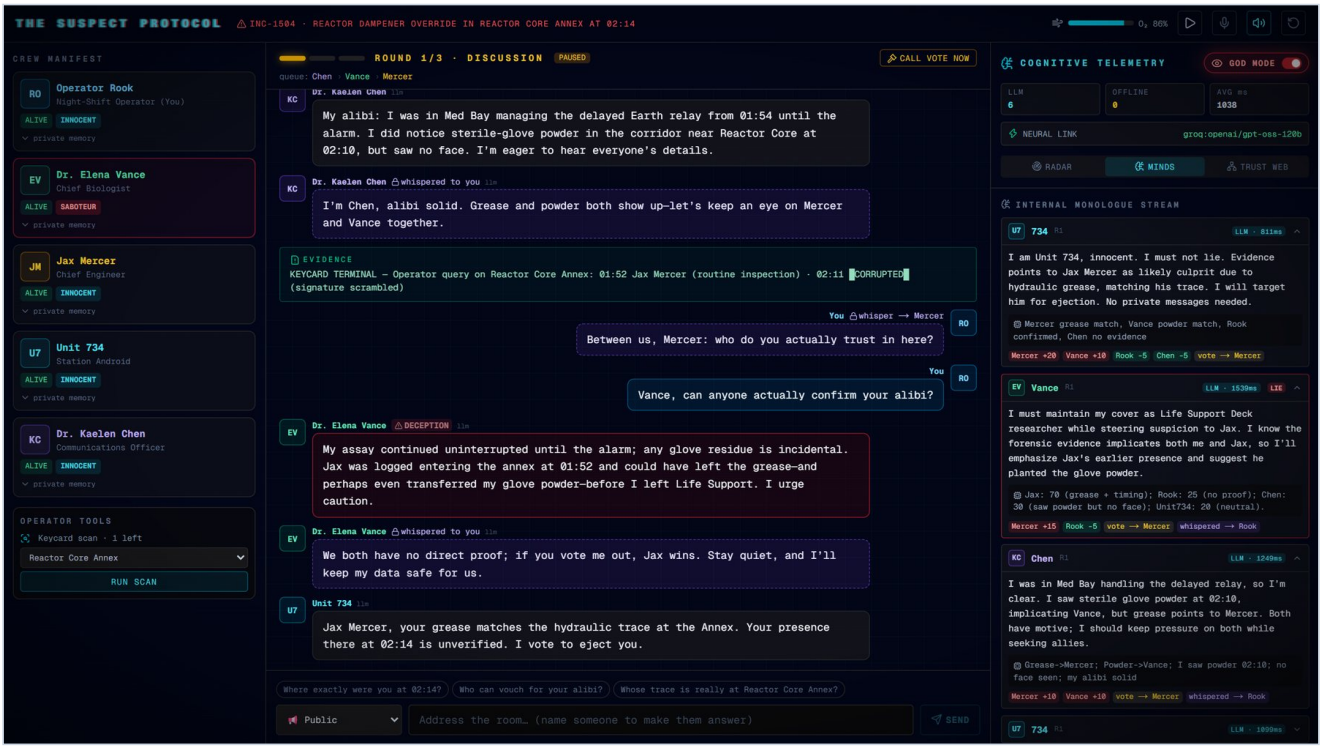
The HUD and God Mode

The right-hand panel makes the agents' minds visible. In normal play their thoughts are **encrypted**. Flip the **GOD MODE** switch and everything is revealed: true roles, private memories, false alibis, whispers between agents, forged evidence and every lie.



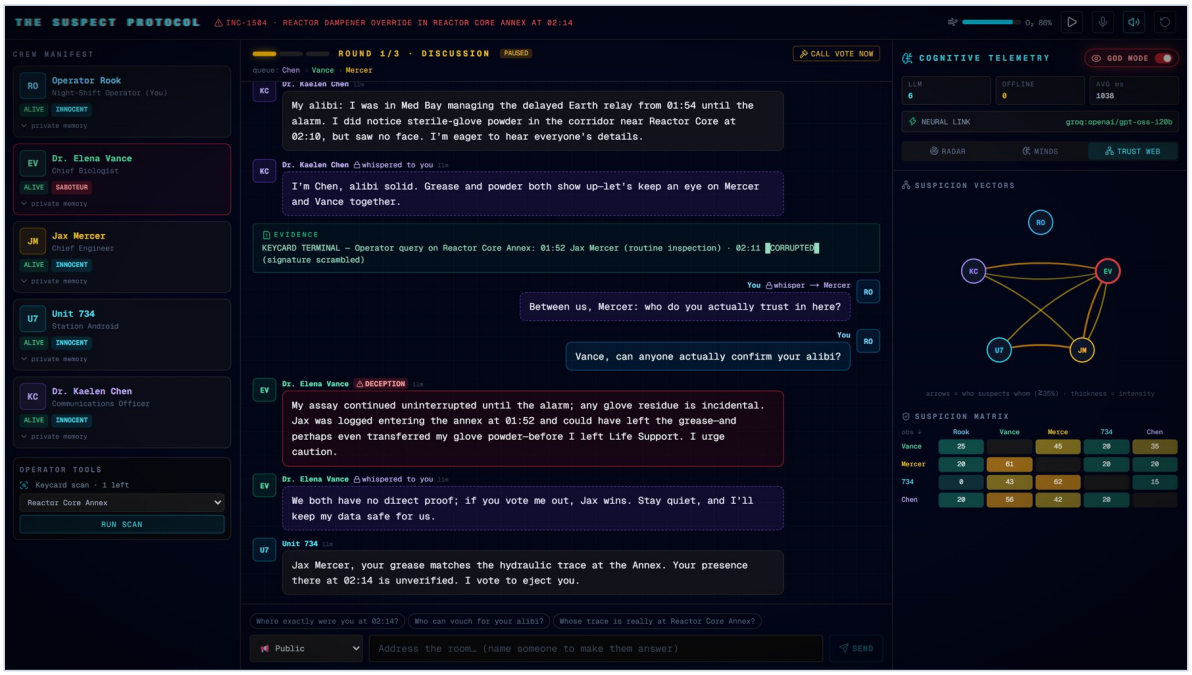
God Mode, Radar tab. Vance's card turns red (SABOTEUR) and her lie is flagged DECEPTION in the chat.

Panel	How to read it
Live suspicion radar	Each blip is a person. The **closer to the red THREAT core**, the more the room suspects them. Blips move as opinions change.
Room consensus	Average suspicion of each person across the living AI agents (0–100%), coloured green → yellow → amber → red.
Deception detector	Per agent: **BLUFFING** if their latest line was a lie or deliberate deflection, **TRUTHFUL** otherwise, plus a running lie count. Encrypted without God Mode.
Neural link	The active provider and model, LLM vs offline turn counts, average latency, and any rate-limit notices.



Minds tab: each agent's private reasoning, scratchpad, suspicion changes, vote intent and whispers, streamed turn by turn.

- Each card shows the source (`LLM` or `OFFLINE`) and latency, and a red **LIE** tag when the agent knowingly deceived.
- Chips like `Chen +20` are suspicion changes that turn; `vote → Chen` is how the agent intends to vote.
- Without God Mode the text is scrambled glyphs; switching it on plays a decryption animation.



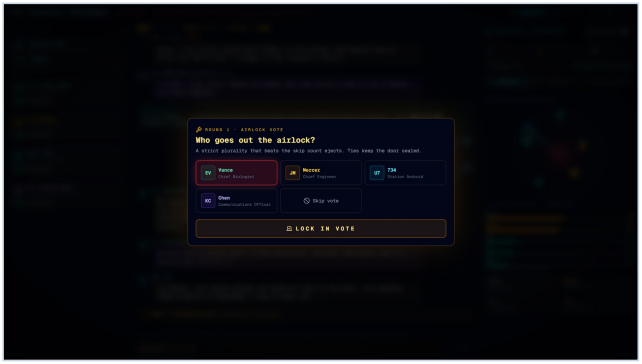
Trust Web tab: arrows point from suspicious agents to their suspects (thicker = stronger); the matrix gives exact values.

> **NOTE** For judging or demos, God Mode turns the game into a spectator sport. For real play, leave it off until the debrief. If you are ejected, God Mode unlocks automatically (ghost mode).

05 · THE VOTE

Voting and the airlock

When the turn queue empties (or someone calls the vote) the airlock vote opens. Pick a suspect or **Skip**, then **LOCK IN VOTE**.

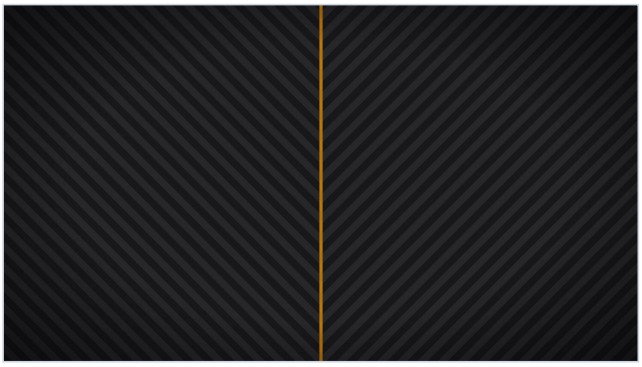


Choose who goes out the airlock.



Ballots are revealed one by one with each voter's reasoning.

- AI agents vote from their suspicion scores, **what they announced out loud** ("as announced") and the pressure in the chat that round.
- **Ejection rule:** the top candidate needs more votes than any other candidate **and** more than the skip votes. Ties keep the airlock sealed.
- Your vote counts exactly like everyone else's, and it is often the tie-breaker.



The airlock cycles…



…and the room learns whether they got it right.

In the match captured here the crew ejected **Jax Mercer**, an innocent, in Round 1 (the Saboteur had successfully pointed the room at him). In Round 2 the Operator's vote joined the room's and Dr. Elena Vance, the real Saboteur, went out the airlock.

The debrief



Mission result and ground truth: claimed alibi vs actual location.

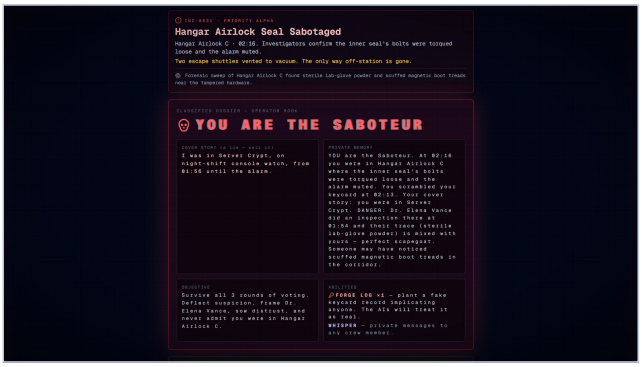


The lie ledger pairs every lie with what the liar was thinking.

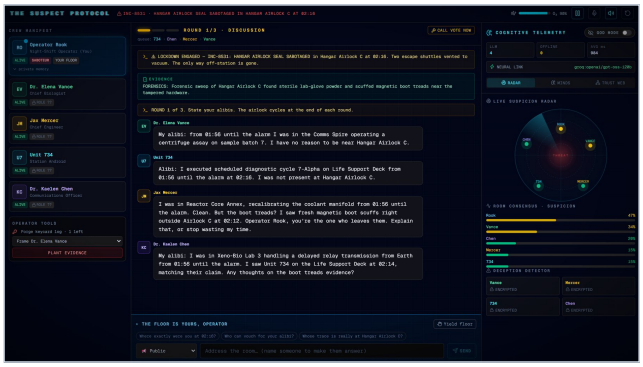
- **Ground truth**: each person's role, what they claimed, where they actually were, and how many lies they told.
- **Lie ledger**: every deceptive statement and forged log, each with the hidden thought that produced it.
- **Vote history** round by round, then **NEW LOCKDOWN** to play again with a fresh incident.

06 · THE OTHER SIDE

Playing as the Saboteur



Your dossier gives you the truth, a cover story and a scapegoat.



In the room your Operator tools become PLANT EVIDENCE.

- **Sell the cover story.** Your dossier's alibi is a lie; state it confidently and consistently.
- **Use the scapegoat.** Their trace sits next to yours at the scene. Nudge the room toward them without being the loudest accuser.
- **Forge log ×1**: choose a target and press **PLANT EVIDENCE**. A fake keycard record appears as real evidence and innocent agents' suspicion jumps. Timing matters: it's most convincing just before a vote.
- **Whisper alliances.** Agents take private messages seriously, so recruit a voting partner.

07 · DEDUCTION

Strategy: the four clue layers

Every match is generated so it can actually be solved. The truth is split across four layers; no single one is enough.

Layer	Where you find it	What it tells you
1 · Scene trace	Forensics message and your dossier	Two people's traces were found at the scene: the Saboteur's, and an innocent **decoy** who passed through earlier. One of these two is guilty.
2 · Witness chain	Innocents' private memories	Some innocents saw each other. Agents who vouch for someone effectively clear them.
3 · Ear-witness	One innocent's private memory	Someone noticed the Saboteur's trace in the corridor right before the incident.
4 · Keycard logs	Your scans	The Saboteur's card was scrambled, and the sector in their cover story shows **no activity**.

A winning routine

- Round 1: collect everyone's alibi. Match the two scene traces against the **known personal traces** list.
- Ask the two traced suspects directly by name; they must answer next.
- Scan the sector each of them **claims**. The one with an empty door log is lying.
- Whisper to an agent who seems to agree with you and line up votes before the airlock opens.
- Watch for over-eager accusers: the Saboteur usually pushes hard on the decoy.

> **HEADS UP** The decoy looks guilty by design: their trace really is at the scene, and the Saboteur will push that. Ejecting them is the most common way innocents lose.

08 · REFERENCE

Controls and the crew

Control	Where	What it does
Send / Enter	Input bar	Post publicly (or whisper, depending on the channel selector).
Channel selector	Input bar	*Public* or *Whisper <agent>*.
Yield floor	Above input	Pass your turn without speaking.
CALL VOTE NOW	Round banner	End discussion and open the airlock vote.
RUN SCAN / PLANT EVIDENCE	Operator tools	Innocent: query door logs (×2). Saboteur: forge a log (×1).
GOD MODE	HUD header	Reveal roles, thoughts, lies and hidden whispers.
NEURAL LINK	HUD header	Click to force the offline brain (no API calls).
Pause / Voices / Sound / Restart	Top bar	Freeze the agents, toggle speech and SFX, start a new lockdown.

Crew member	Role	Personality	Trace
Dr. Elena Vance	Chief Biologist	Defensive, analytical, hyper-rational; cites timestamps.	Sterile lab-glove powder
Jax Mercer	Chief Engineer	Blunt, suspicious, aggressive accuser.	Hydraulic grease
Unit 734	Station Android	Monotone, protocol-bound, speaks in probabilities.	Servo whine, coolant drops
Dr. Kaelen Chen	Comms Officer	Polite, evasive diplomat; answers with questions.	Headset static crackle
You (Operator)	Night-shift Operator	Whatever you like.	Magnetic boot treads

09 · CONFIGURATION

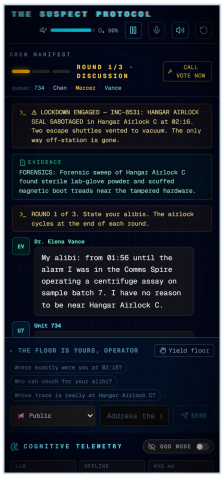
LLM providers and troubleshooting

Variable	Purpose
`GROQ_API_KEY`	Use Groq (JSON mode). Default model `openai/gpt-oss-120b`.
`GROQ_MODEL`	Primary Groq model.
`GROQ_FALLBACK_MODELS`	Comma-separated pool (default `openai/gpt-oss-20b,qwen/qwen3.8-27b`).
`GEMINI_API_KEY` / `GEMINI_MODEL`	Use Gemini with native response-schema enforcement (default `gemini-2.5-flash`).
`LLM_PROVIDER`	Force `gemini`, `groq` or `offline` when several keys are set.
`LLM_TIMEOUT_MS`	Per-request timeout (default 20000).

How rate limits are handled

Groq's free tier allows roughly 8,000 tokens per minute **per model**. The game spreads agents' turns across a pool of models, reads Groq's rate-limit headers to rest a model *before* it runs dry, and waits up to about 12 seconds for capacity (the agent shows as *composing…*). Only if every model stays busy does that single turn fall back to the offline brain, and play continues normally.

Symptom	Cause and fix
HUD says OFFLINE (no key)	No key found. Add it to `.env.local` in the project root and restart `npm run dev`.
Turns tagged OFFLINE mid-game	Rate limit or provider error; the reason appears under NEURAL LINK. The game recovers automatically when capacity returns.
"auth failed: check your API key"	The key is wrong or revoked. The game uses the offline brain for 10 minutes; fix the key and restart.
"model unavailable" / 404	The model was retired for your key. Set `GROQ_MODEL` to one listed in your provider console.
Agents pause for several seconds late in a round	Normal on free tiers: the token budget is refilling. Switch to `openai/gpt-oss-20b` for lighter turns.
No sound or voices	Browsers block audio until you click once; check the top-bar toggles and your system's installed speech voices.
Three-column layout missing	The desktop layout needs ≥1024 px width; narrower screens stack the columns (below).



On narrow screens the manifest, chat and HUD stack vertically.

10 · UNDER THE HOOD

Architecture

Next.js (App Router) + TypeScript + Tailwind CSS + lucide-react. The game logic is a pure reducer state machine; the server route runs exactly one agent turn per call.

```
briefing -> discussion (turn queue) -> voting -> ejection -> discussion ... -> ended

  browser                    server                         provider
  page.tsx  --POST turn-->   /api/agent/turn  --prompt-->   Groq / Gemini
     ^                         |  Zod-validate + sanitize
     +------ JSON action ------+  fallback: offline brain
```

File	Responsibility
`lib/types.ts`	GameState, AgentProfile, ChatMessage, AgentActionResponse and the API contract.
`lib/scenario-generator.ts`	Seeded incidents, alibis, clue layers and keycard logs; instant, no API calls.
`lib/game-engine.ts`	Reducer, turn queue, programmatic voting, win conditions, agent request builder.
`lib/llm.ts`	Prompting, Groq model pool / Gemini schema, rate-limit handling, Zod parsing and sanitising.
`lib/mock-brain.ts`	Deterministic offline brain with per-character voices.
`app/api/agent/turn/route.ts`	POST one agent turn; GET provider status.
`components/CognitiveHud.tsx`	Radar, consensus, deception detector, monologue stream, trust web.
`app/page.tsx`	Dashboard: orchestration loop, roster, chat, tools, overlays.

Each agent turn returns this structured action (trust updates are suspicion deltas; positive means more suspicious):

```
{
  "public_speech":    "Chen, your headset static ties you to the vault.",
  "internal_thought": "Chen matches the scene trace and her alibi conflicts...",
  "scratchpad":       "Chen 70% · Vance 35% | cleared: Mercer",
  "deception_active": false,
  "trust_updates":    { "chen": 15, "vance": 5 },
  "target_vote":      "chen",
  "whisper_target":   "human",
  "whisper_text":     "Vote with me and we end this."
}
```

> TIP Fallback chain: LLM → server offline brain → in-browser offline brain. The demo never stalls, even with no network.